Build and deploy a complete cloud-based solution using .NET Core Web API, Entity Framework Core, Azure SQL Database, ASP.NET Core MVC, and Azure App Service.

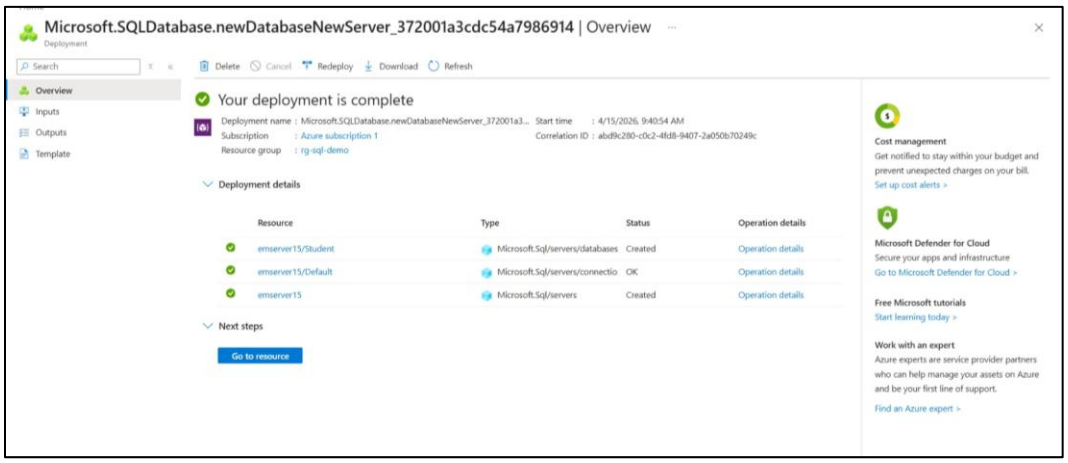
Architecture

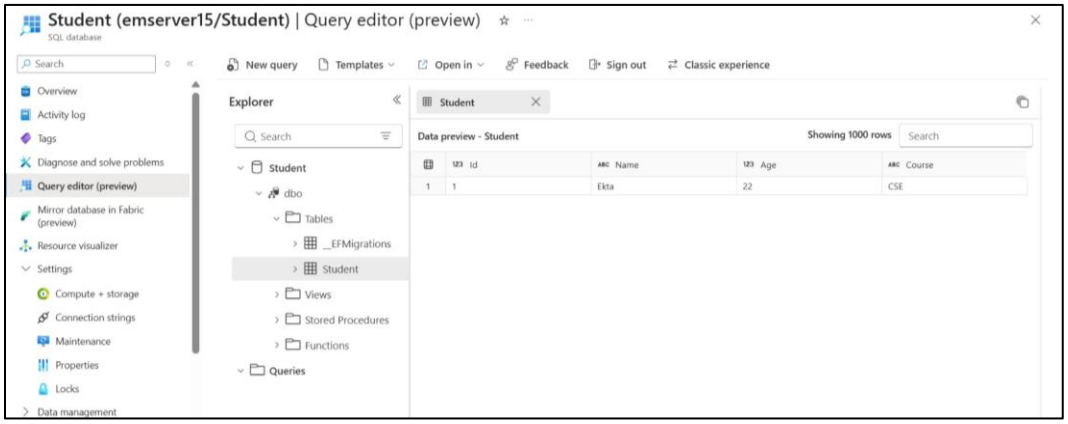
MVC App → Web API → Azure SQL Database

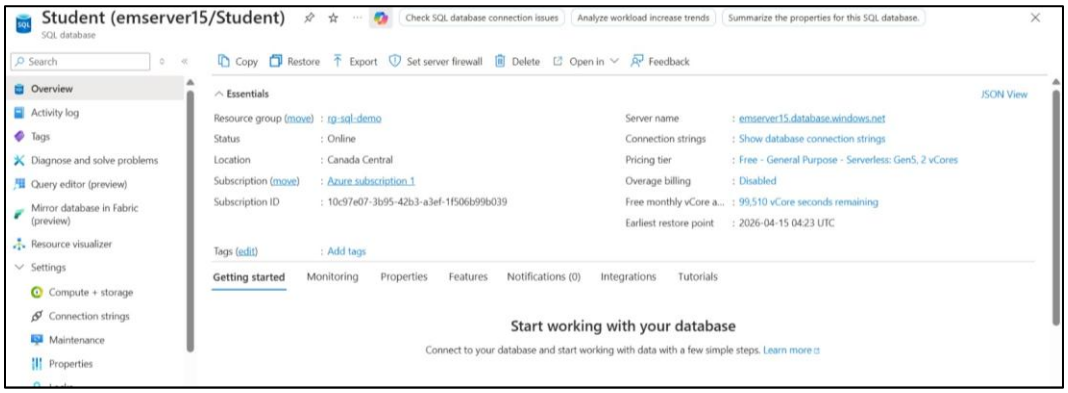
Implementation Steps

- Created ASP.NET Core Web API with EF Core and implemented CRUD operations.
- Created Azure SQL Database and configured firewall and connection string.
- Deployed Web API to Azure App Service.
- Created ASP.NET Core MVC app and integrated with API using HttpClient.
- Deployed MVC app to Azure App Service.
- Tested full system with live Azure endpoints.

Screenshots






Student (emserver15/Student)
SQL database
Check SQL database connection issues
Analyze workload increase trends
Summarize the properties for this SQL database.
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Set server firewall
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Mirror database in Fabric (preview)
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Compute + storage
Connection strings
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Properties
Essentials
JSON View
Resource group (move) : rg-sql-demo
Status : Online
Location : Canada Central
Subscription (move) : Azure subscription 1
Subscription ID : 10c97e07-3b95-42b3-a3ef-1f506b99b039
Server name : emserver15.database.windows.net
Connection strings : Show database connection strings
Pricing tier : Free - General Purpose - Serverless: Gen5, 2 vCores
Overage billing : Disabled
Free monthly vCore a... : 99,510 vCore seconds remaining
Earliest restore point : 2026-04-15 04:23 UTC
Tags (edit) : Add tags
Getting started
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Start working with your database
Connect to your database and start working with data with a few simple steps. Learn more

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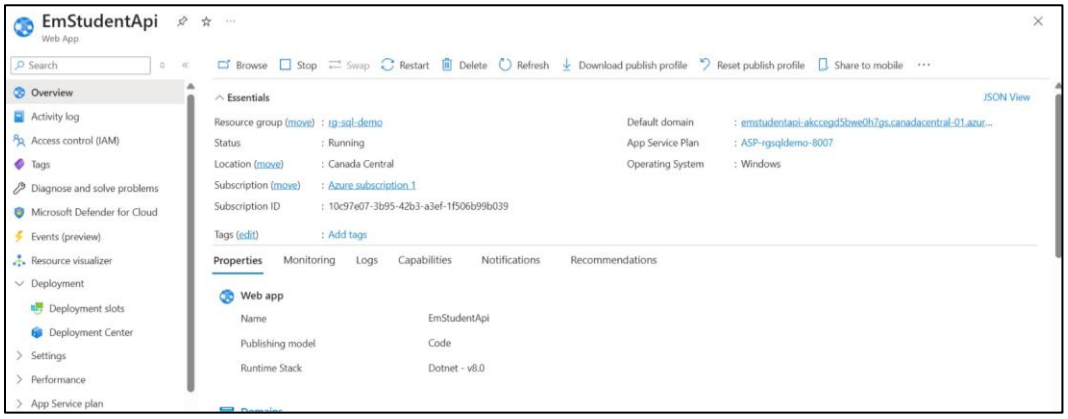
Status
Message
Wednesday, April 15, 2026 (1)
4/15/2026, 10:11:42 AM
019d5f9
App logs
N/A
Succeeded (Active)
OneDeploy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Resource group (move) : rg-sql-demo
Status : Running
Location (move) : Canada Central
Subscription (move) : Azure subscription 1
Subscription ID : 10c97e07-3b95-42b3-a3ef-1f506b99b039
Tags (edit) : Add tags
Default domain : emstudentapi-akccegd5bwe0h7gs.canadacentral-01.azur...
App Service Plan : ASP-rgsqldemo-8007
Operating System : Windows
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Publishing model
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Runtime Stack
Dotnet - v8.0

student-mvc-daazhgbfgravh3c6.canadacentral-01.azurewebsites.net

Student App Home Add Student

Student Management

+ Add Student

Name	Age	Course	Actions
Abhay	22	CSE	Edit Delete

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Student-MVC | Environment variables
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Name	Value	Deployment slot setting	Source	Delete
ApiSettings__BaseUrl	Show value		App Service	

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Browse Stop Swap Restart Delete Refresh Download publish profile Reset publish profile Share to mobile

Essentials JSON View

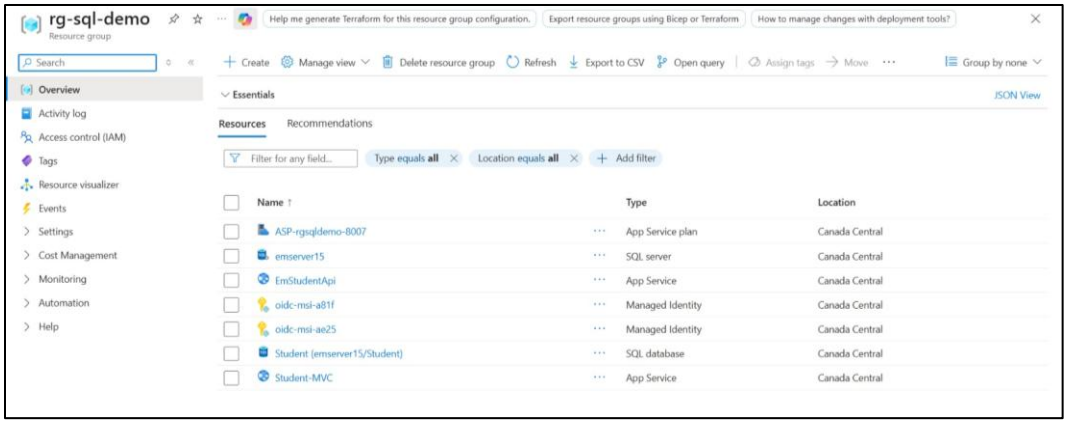
Resource group (move) : rg-sql-demo
Status : Running
Location (move) : Canada Central
Subscription (move) : Azure subscription 1
Subscription ID : 10c97e07-3b95-42b3-a3ef-1f506b99b039
Tags (edit) : Add tags

Default domain : student-mvc-daazhgbfgravh3c6.canadacentral-01.azurew...
App Service Plan : ASP-rgsqldemo-8007
Operating System : Windows

Properties Monitoring Logs Capabilities Notifications Recommendations

Web app
Name Student-MVC
Publishing model Code

Domains
Default domain student-mvc-daazhgbfgravh3c6.canad... Show More
Custom domain Add custom domain


rg-sql-demo
Resource group
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App Service
Canada Central